

Fragile Watermarking: Visual Evaluation Report

Generated: May 04, 2026 18:54 | Total Experiments: 32

Experiment 1: Peppers — Attack: Crop 33%



1. Original

2. Watermarked

3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	36.03 dB
Watermarked SSIM	0.9972	Recovered SSIM	0.9723
True Positive Rate (TPR)	1.0000	Recovered NC	0.9995
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

Experiment 2: Peppers — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	38.53 dB
Watermarked SSIM	0.9972	Recovered SSIM	0.9842
True Positive Rate (TPR)	1.0000	Recovered NC	0.9997
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

Experiment 3: Peppers — Attack: Salt_and_Pepper_1%



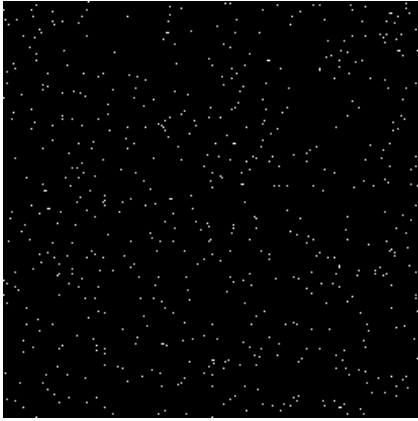
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	29.34 dB
Watermarked SSIM	0.9972	Recovered SSIM	0.8993
True Positive Rate (TPR)	0.5489	Recovered NC	0.9978
False Positive Rate (FPR)	0.0389	Tamper F1-Score	0.1914

Experiment 4: Peppers — Attack: JPEG_80



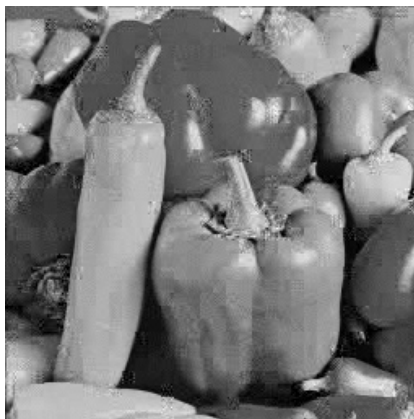
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	27.90 dB
Watermarked SSIM	0.9972	Recovered SSIM	0.8023
True Positive Rate (TPR)	1.0000	Recovered NC	0.9970
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

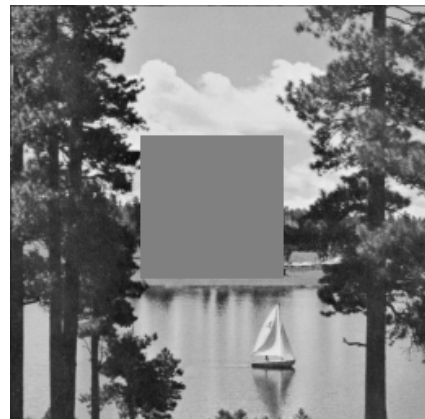
Experiment 5: Sailboat on lake — Attack: Crop 33%



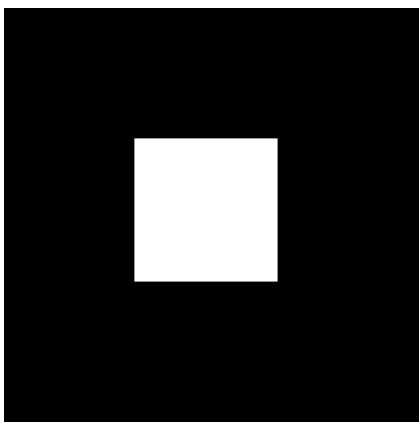
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	32.67 dB
Watermarked SSIM	0.9976	Recovered SSIM	0.9581
True Positive Rate (TPR)	1.0000	Recovered NC	0.9991
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

Experiment 6: Sailboat on lake — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



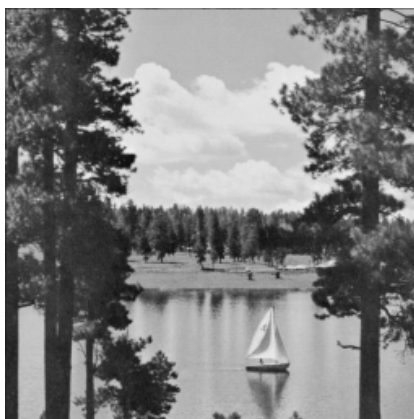
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	37.51 dB
Watermarked SSIM	0.9976	Recovered SSIM	0.9826
True Positive Rate (TPR)	1.0000	Recovered NC	0.9997
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

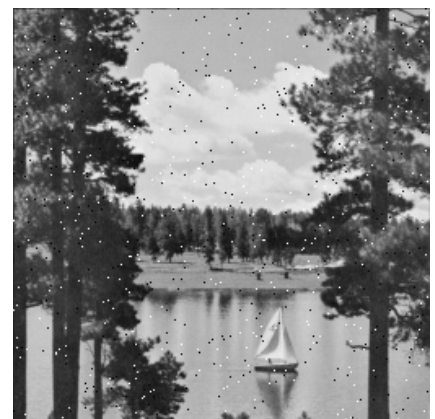
Experiment 7: Sailboat on lake — Attack: Salt_and_Pepper_1%



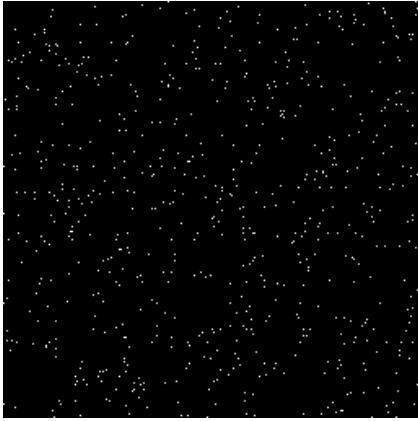
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	27.91 dB
Watermarked SSIM	0.9976	Recovered SSIM	0.8951
True Positive Rate (TPR)	0.5800	Recovered NC	0.9974
False Positive Rate (FPR)	0.0461	Tamper F1-Score	0.1917

Experiment 8: Sailboat on lake — Attack: JPEG_80



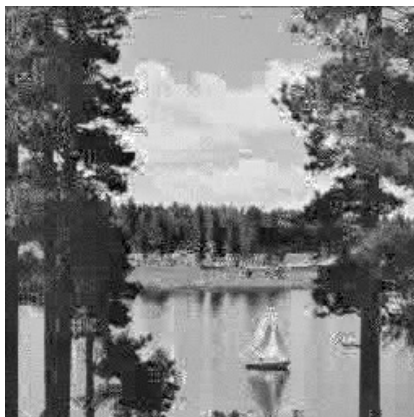
1. Original



2. Watermarked



3. Attacked

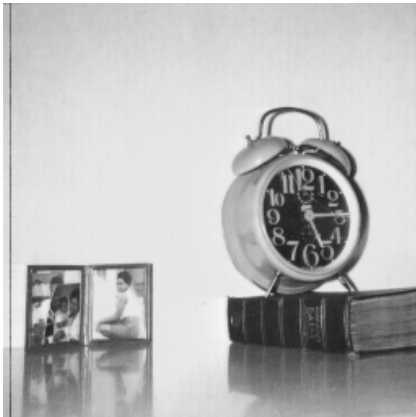


4. True Tamper Map

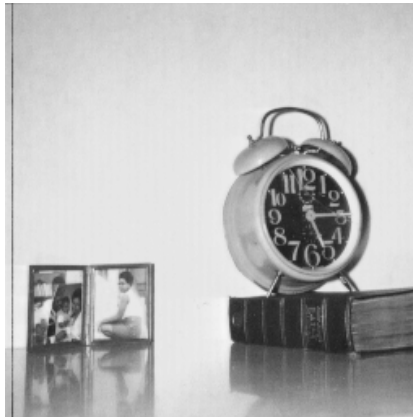
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	23.37 dB
Watermarked SSIM	0.9976	Recovered SSIM	0.7041
True Positive Rate (TPR)	1.0000	Recovered NC	0.9925
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

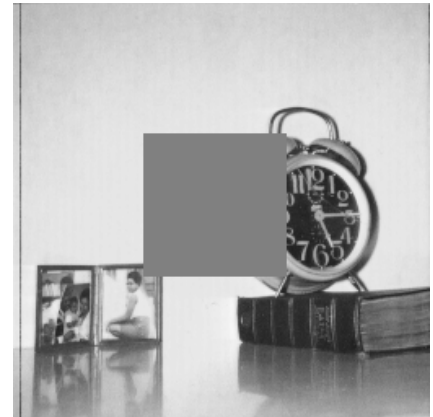
Experiment 9: Clock — Attack: Crop 33%



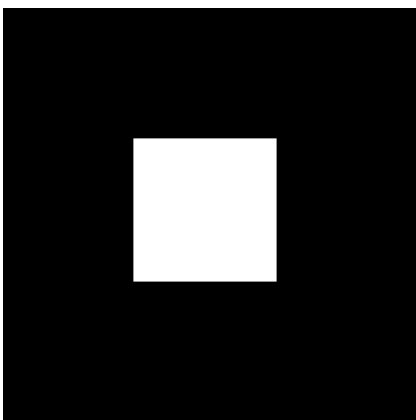
1. Original



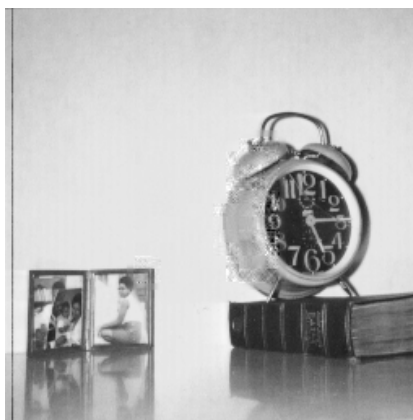
2. Watermarked



3. Attacked



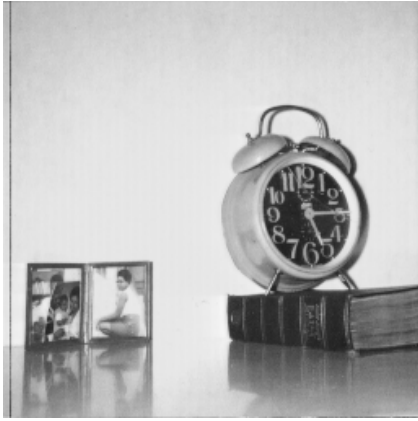
4. True Tamper Map



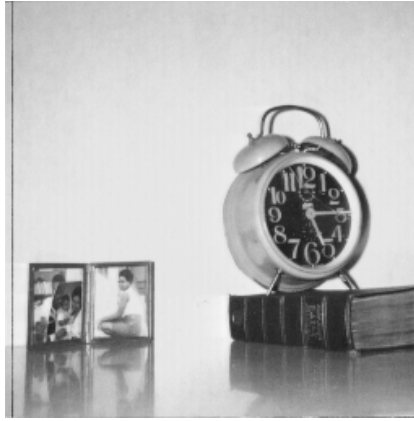
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	37.33 dB
Watermarked SSIM	0.9951	Recovered SSIM	0.9806
True Positive Rate (TPR)	1.0000	Recovered NC	0.9998
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

Experiment 10: Clock — Attack: Copy-Move



1. Original



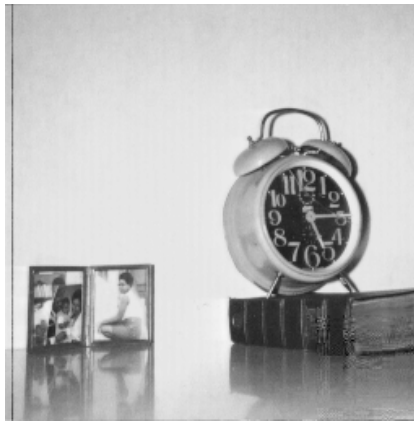
2. Watermarked



3. Attacked



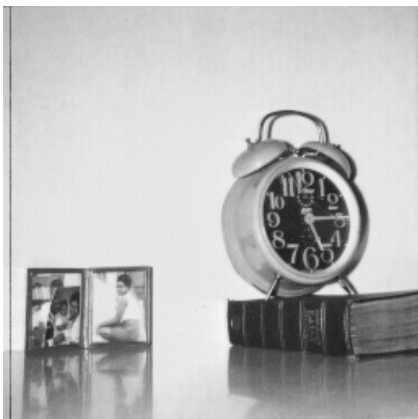
4. True Tamper Map



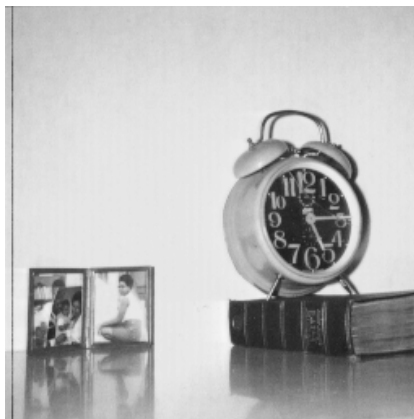
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	39.44 dB
Watermarked SSIM	0.9951	Recovered SSIM	0.9790
True Positive Rate (TPR)	1.0000	Recovered NC	0.9999
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

Experiment 11: Clock — Attack: Salt_and_Pepper_1%



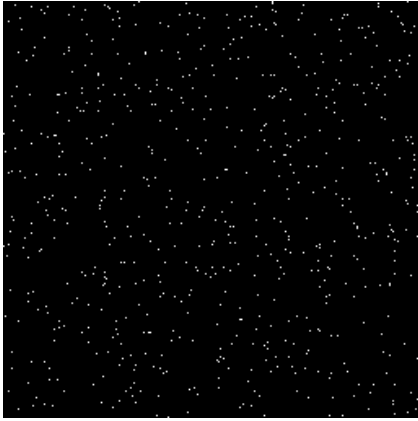
1. Original



2. Watermarked



3. Attacked



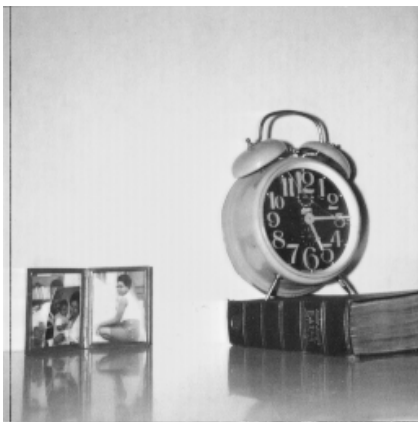
4. True Tamper Map



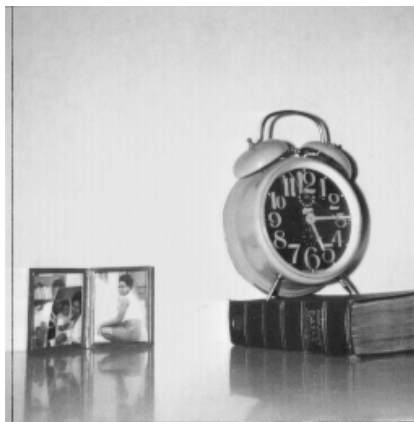
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	28.93 dB
Watermarked SSIM	0.9951	Recovered SSIM	0.9007
True Positive Rate (TPR)	0.5730	Recovered NC	0.9989
False Positive Rate (FPR)	0.0433	Tamper F1-Score	0.1931

Experiment 12: Clock — Attack: JPEG_80



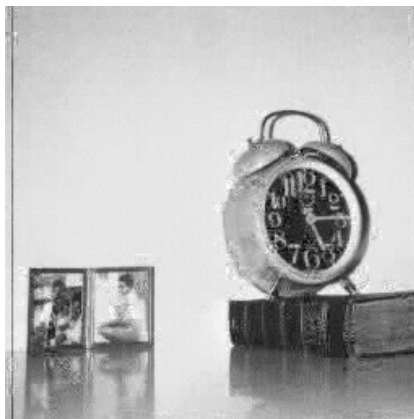
1. Original



2. Watermarked



3. Attacked

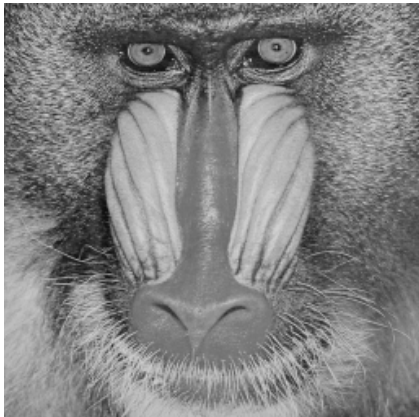


4. True Tamper Map

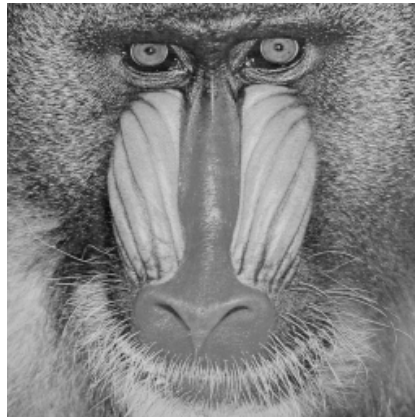
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	30.27 dB
Watermarked SSIM	0.9951	Recovered SSIM	0.8933
True Positive Rate (TPR)	1.0000	Recovered NC	0.9992
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

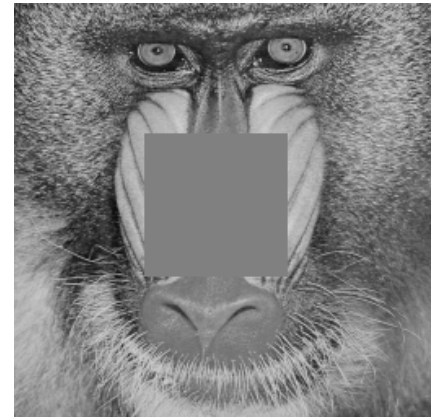
Experiment 13: Mandrill — Attack: Crop 33%



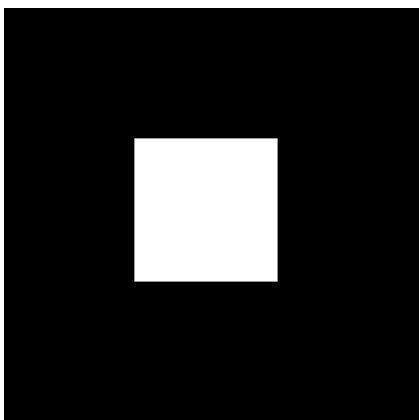
1. Original



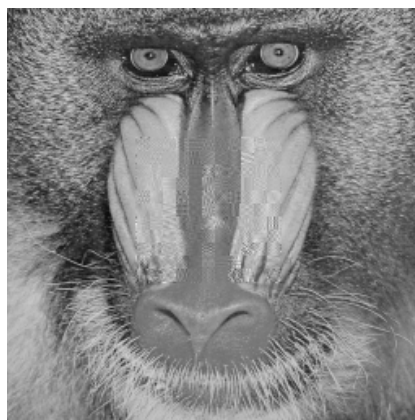
2. Watermarked



3. Attacked



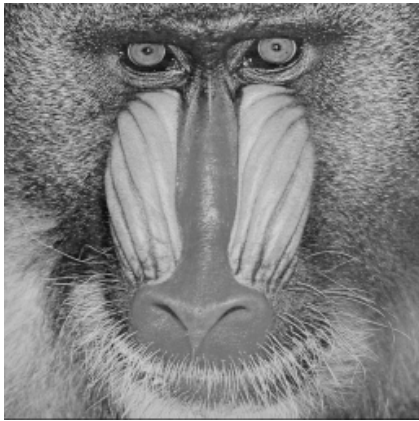
4. True Tamper Map



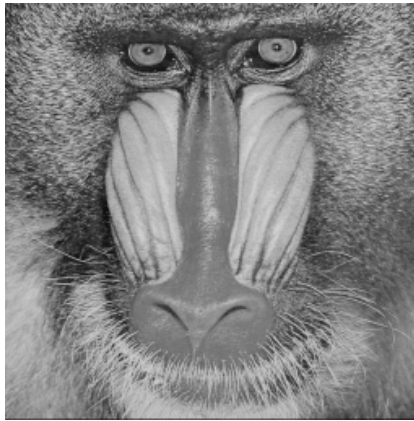
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	34.21 dB
Watermarked SSIM	0.9989	Recovered SSIM	0.9449
True Positive Rate (TPR)	1.0000	Recovered NC	0.9993
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

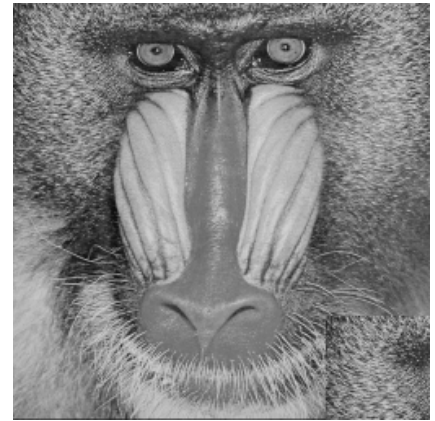
Experiment 14: Mandrill — Attack: Copy-Move



1. Original



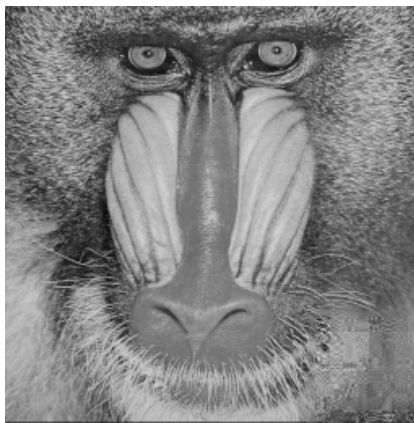
2. Watermarked



3. Attacked



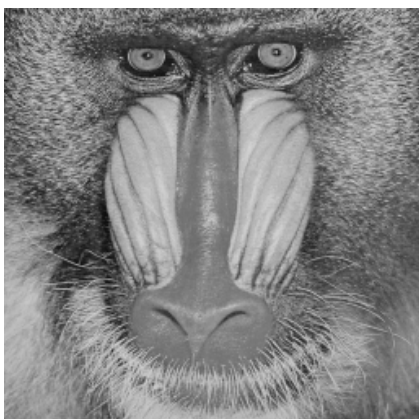
4. True Tamper Map



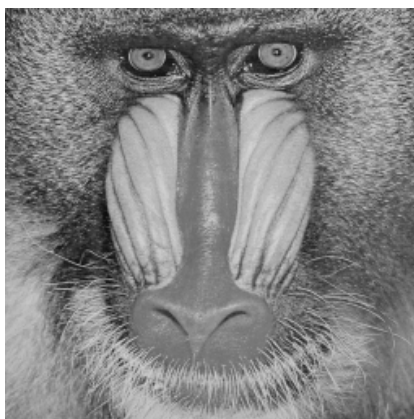
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	34.85 dB
Watermarked SSIM	0.9989	Recovered SSIM	0.9640
True Positive Rate (TPR)	1.0000	Recovered NC	0.9994
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

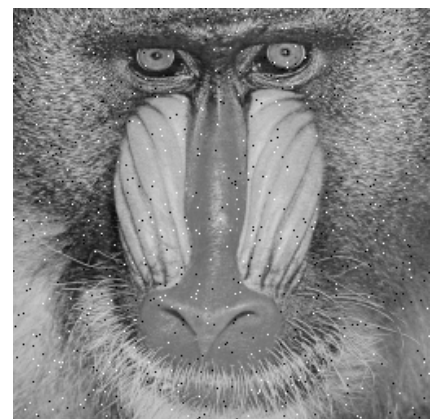
Experiment 15: Mandrill — Attack: Salt_and_Pepper_1%



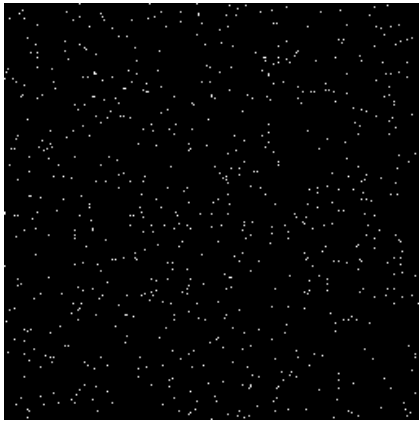
1. Original



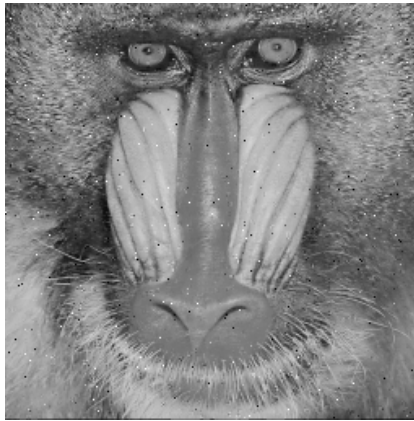
2. Watermarked



3. Attacked



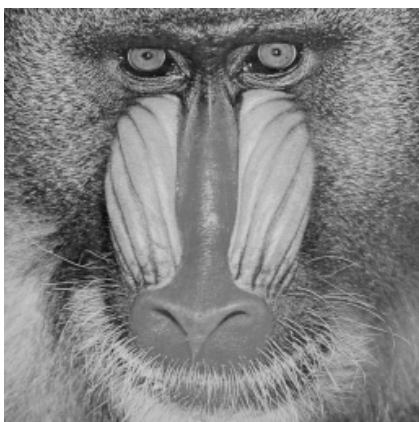
4. True Tamper Map



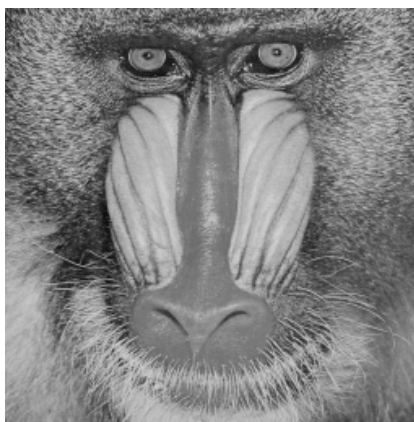
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	27.46 dB
Watermarked SSIM	0.9989	Recovered SSIM	0.8728
True Positive Rate (TPR)	0.5430	Recovered NC	0.9968
False Positive Rate (FPR)	0.0439	Tamper F1-Score	0.1931

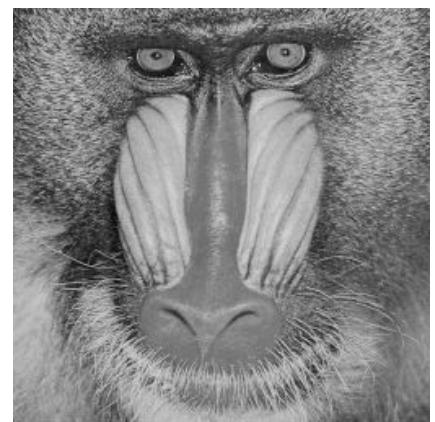
Experiment 16: Mandrill — Attack: JPEG_80



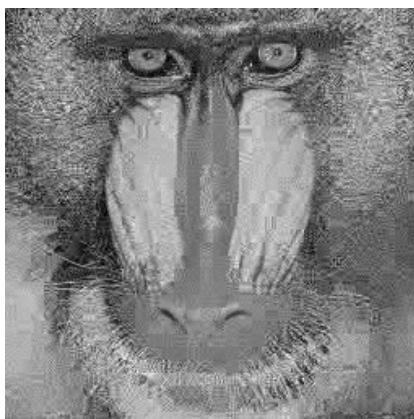
1. Original



2. Watermarked



3. Attacked

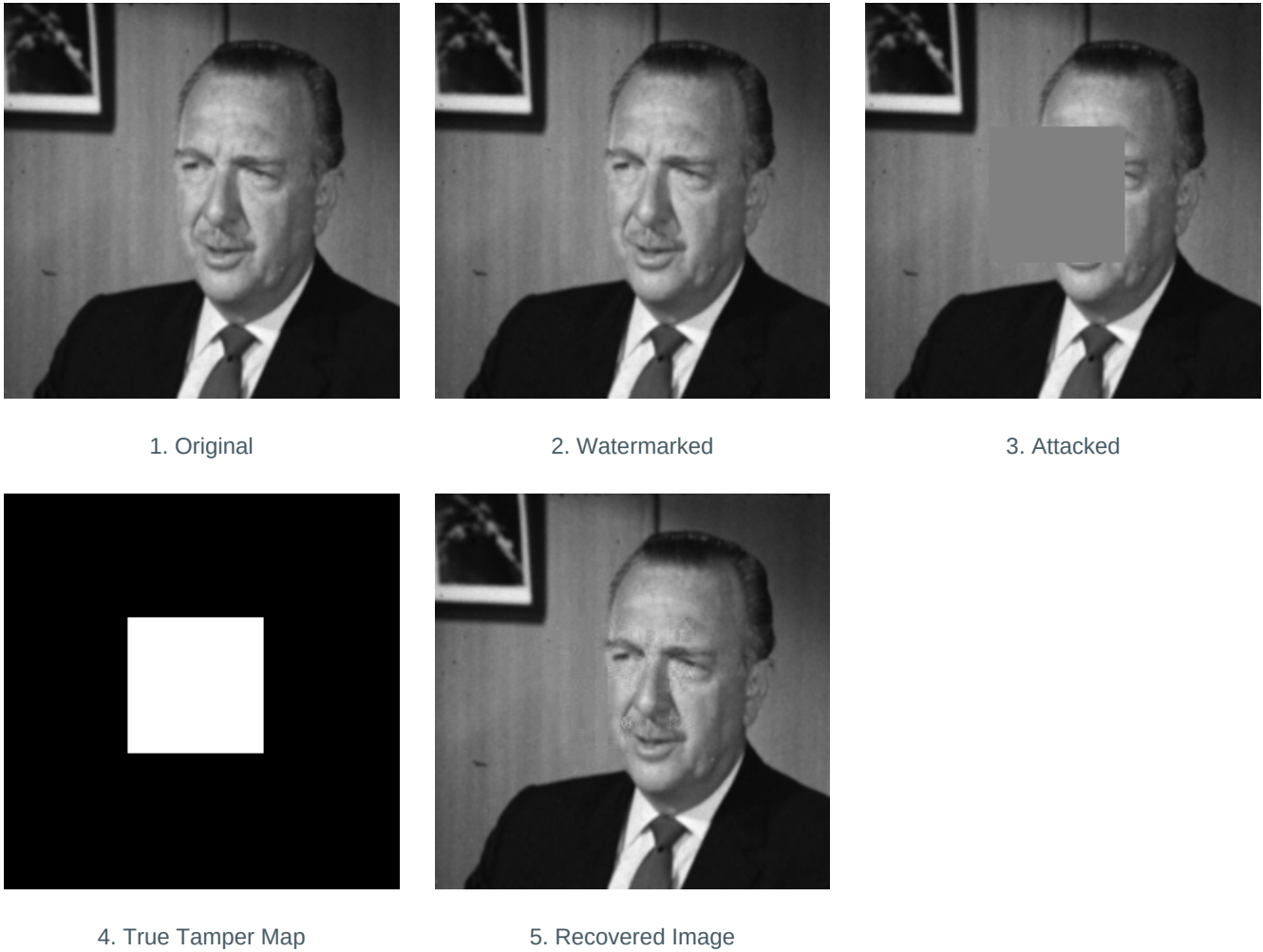


5. Recovered Image

4. True Tamper Map

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	21.51 dB
Watermarked SSIM	0.9989	Recovered SSIM	0.5413
True Positive Rate (TPR)	1.0000	Recovered NC	0.9874
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

Experiment 17: Walter Cronkite — Attack: Crop 33%



Metric	Value	Metric	Value
Watermarked PSNR	51.10 dB	Recovered PSNR	42.62 dB
Watermarked SSIM	0.9953	Recovered SSIM	0.9821
True Positive Rate (TPR)	1.0000	Recovered NC	0.9998
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

Experiment 18: Walter Cronkite — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.10 dB	Recovered PSNR	48.79 dB
Watermarked SSIM	0.9953	Recovered SSIM	0.9938
True Positive Rate (TPR)	1.0000	Recovered NC	1.0000
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

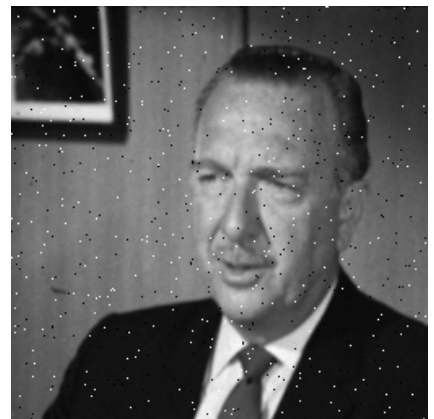
Experiment 19: Walter Cronkite — Attack: Salt_and_Pepper_1%



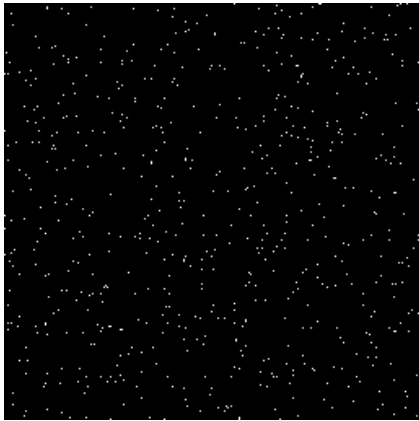
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.10 dB	Recovered PSNR	30.55 dB
Watermarked SSIM	0.9953	Recovered SSIM	0.9038
True Positive Rate (TPR)	0.5919	Recovered NC	0.9973
False Positive Rate (FPR)	0.0449	Tamper F1-Score	0.1930

Experiment 20: Walter Cronkite — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.10 dB	Recovered PSNR	36.55 dB
Watermarked SSIM	0.9953	Recovered SSIM	0.9177
True Positive Rate (TPR)	0.9999	Recovered NC	0.9993
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

Experiment 21: Chemical plant — Attack: Crop 33%



1. Original

2. Watermarked

3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	32.98 dB
Watermarked SSIM	0.9986	Recovered SSIM	0.9545
True Positive Rate (TPR)	1.0000	Recovered NC	0.9987
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

Experiment 22: Chemical plant — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	36.03 dB
Watermarked SSIM	0.9986	Recovered SSIM	0.9825
True Positive Rate (TPR)	1.0000	Recovered NC	0.9994
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

Experiment 23: Chemical plant — Attack: Salt_and_Pepper_1%



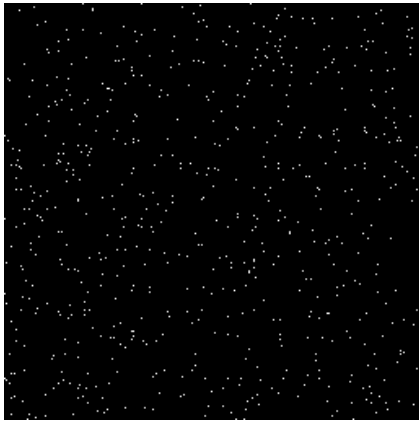
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	28.60 dB
Watermarked SSIM	0.9986	Recovered SSIM	0.8966
True Positive Rate (TPR)	0.5802	Recovered NC	0.9965
False Positive Rate (FPR)	0.0452	Tamper F1-Score	0.1902

Experiment 24: Chemical plant — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	24.70 dB
Watermarked SSIM	0.9986	Recovered SSIM	0.6762
True Positive Rate (TPR)	1.0000	Recovered NC	0.9913
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

Experiment 25: House — Attack: Crop 33%



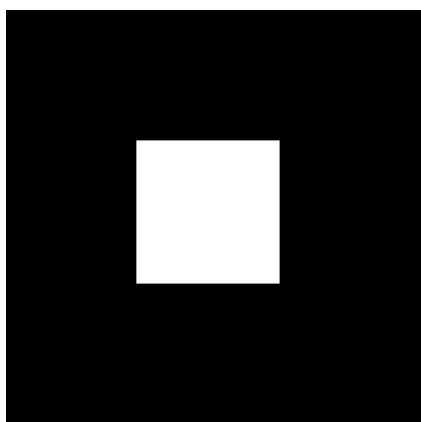
1. Original



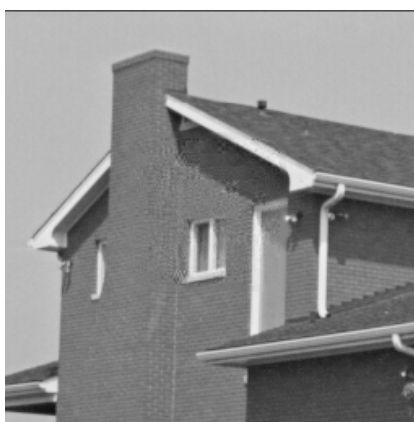
2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	39.05 dB
Watermarked SSIM	0.9958	Recovered SSIM	0.9637
True Positive Rate (TPR)	1.0000	Recovered NC	0.9998
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

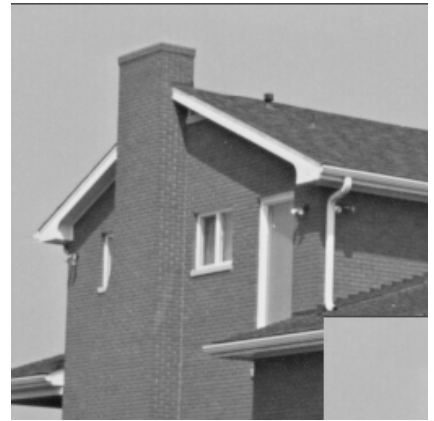
Experiment 26: House — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	40.27 dB
Watermarked SSIM	0.9958	Recovered SSIM	0.9825
True Positive Rate (TPR)	1.0000	Recovered NC	0.9999
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

Experiment 27: House — Attack: Salt_and_Pepper_1%



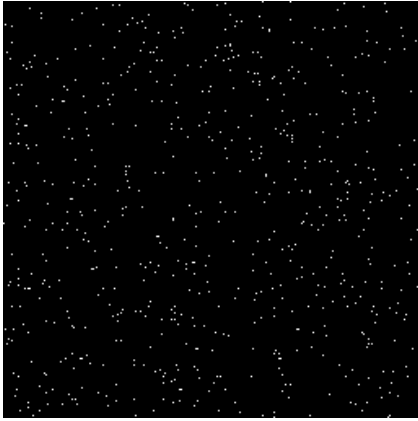
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	29.56 dB
Watermarked SSIM	0.9958	Recovered SSIM	0.8771
True Positive Rate (TPR)	0.5590	Recovered NC	0.9983
False Positive Rate (FPR)	0.0428	Tamper F1-Score	0.1924

Experiment 28: House — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	31.29 dB
Watermarked SSIM	0.9958	Recovered SSIM	0.8260
True Positive Rate (TPR)	1.0000	Recovered NC	0.9989
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

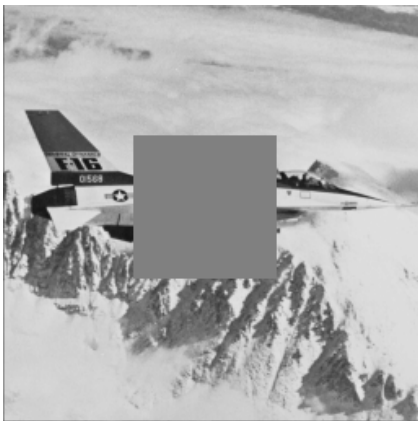
Experiment 29: Airplane — Attack: Crop 33%



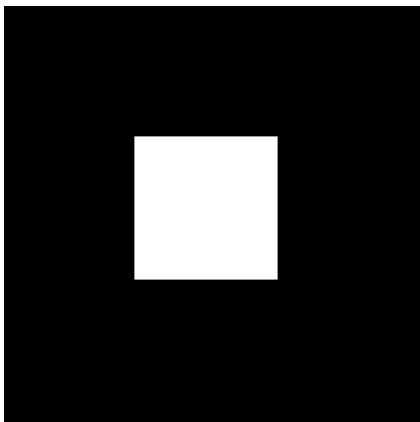
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.11 dB	Recovered PSNR	32.89 dB
Watermarked SSIM	0.9963	Recovered SSIM	0.9701
True Positive Rate (TPR)	1.0000	Recovered NC	0.9995
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

Experiment 30: Airplane — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.11 dB	Recovered PSNR	35.45 dB
Watermarked SSIM	0.9963	Recovered SSIM	0.9794
True Positive Rate (TPR)	1.0000	Recovered NC	0.9997
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

Experiment 31: Airplane — Attack: Salt_and_Pepper_1%



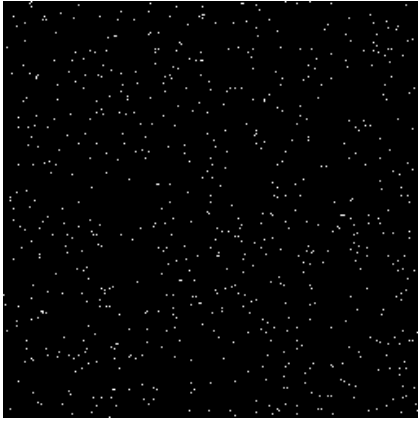
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.11 dB	Recovered PSNR	28.60 dB
Watermarked SSIM	0.9963	Recovered SSIM	0.8859
True Positive Rate (TPR)	0.5901	Recovered NC	0.9987
False Positive Rate (FPR)	0.0480	Tamper F1-Score	0.1931

Experiment 32: Airplane — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.11 dB	Recovered PSNR	25.90 dB
Watermarked SSIM	0.9963	Recovered SSIM	0.7909
True Positive Rate (TPR)	1.0000	Recovered NC	0.9976
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000
